

Restrained page-local apparatus

page-local note; exact one-page correspondence to authority physical 1

Authority physical 1 / printed 80. The title-page pixels and objects establish “Exposé V,” the exact French title, the principal byline “par *P. DELIGNE*,” and the separate credit “avec la collaboration de *L. ILLUSIE* (*).” The summary, scope sentence, rule, and C.N.R.S. footnote are work matter and are retained. No printed folio is visible. A newly extracted 1920 × 2850 one-bit full-page authority image is the diplomatic layout control; its lossless PNG presentation derivative is pixel-identical. No inherited byte was accepted.

Restrained page-local apparatus

page-local note; exact one-page correspondence to authority physical 2

Authority physical 2 / printed 81. Folio 81 is recorded here and omitted from reader prose. The page contains the opening of the Introduction, the p -divisible-group and Ext^1 descriptions, and displays (0.1)–(0.2). The comma after “ $d \log q$ ” and the continuation onto physical 3 are preserved. The authority full-page facsimile is the layout and mathematical-symbol control. No inherited byte was accepted.

Restrained page-local apparatus

page-local note; exact one-page correspondence to authority physical 3

Authority physical 3 / printed 82. Folio 82 is recorded here and omitted from reader prose. Display (0.3), the ordinary K_3 definition, the twenty canonical coordinates, and the crystalline Chern-class statement were checked against authority pixels. The page ends mid-sentence with “du”; no closure was supplied. No inherited byte was accepted.

Restrained page-local apparatus

page-local note; exact one-page correspondence to authority physical 4

Authority physical 4 / printed 83. Folio 83 is recorded here and omitted from reader prose. The Introduction closes, Section 1 and 1.1.1 open, and the displayed connection and PD-differential formulas are retained. The condition " $i \neq j$ " and the terminal word "la" were checked in the authority image. No inherited byte was accepted.

Restrained page-local apparatus

page-local note; exact one-page correspondence to authority physical 5

Authority physical 5 / printed 84. Folio 84 is recorded here and omitted from reader prose. Lemma 1.1.2, its underlined statement, the two-arrow composite, formula (1.1.2.1), divided-power notation, and the footnote are present. The page ends with “Nous noterons,” continuing on physical 6. No inherited byte was accepted.

Restrained page-local apparatus

page-local note; exact one-page correspondence to authority physical 6

Authority physical 6 / printed 85. Folio 85 is recorded here and omitted from reader prose. The completed tensor product, the PD-envelope construction, the source’s exact “ d_1^*M / d_0^*M ” notation, the θ – formula marked (*), and the commutative diagram were retained without silent correction. The page ends after the diagram with a comma and continues on physical 7. No inherited byte was accepted.

Restrained page-local apparatus

page-local note; exact one-page correspondence to authority physical 7

Authority physical 7 / printed 86. Folio 86 is recorded here and omitted from reader prose. The page completes the proof of Lemma 1.1.2, then opens 1.1.3. The authority triangle (1.1.3.1) has $F(\varphi)$ horizontal, $\chi(\varphi, \psi)$ vertical, and $F(\psi)$ diagonal; the horizontality square (1.1.3.2) retains the source label $1 \otimes F(\varphi)$. The page ends before the next commutative square, after the word “carré.” No inherited byte was accepted.

Restrained page-local apparatus

page-local note; exact one-page correspondence to authority physical 8

Authority physical 8 / printed 87. Folio 87 is recorded here and omitted from reader prose. The opening square completes the sentence and diagram begun on physical 7. Formulas (1.1.3.3) and (1.1.3.4), including the divided-power summation and the source assignment $f = \psi$, $g = \varphi$, were checked against authority pixels. Sections 1.1.4 and 1.1.5 complete the Teichmüller-lifting and Newton/Hodge-polygon review. No inherited byte was accepted.

Restrained page-local apparatus

page-local note; exact one-page correspondence to authority physical 9

Authority physical 9 / printed 88. Folio 88 is recorded here and omitted from reader prose. Section 1.2 and Proposition 1.2.2 open; the underlined result statement, $\varphi^*(\Omega^1_{A/W}) \subset p\Omega^1_{A/W}$, the two preliminary remarks, and the choice $t_j \mapsto t_j^P$ are retained. The proof ends after “pour” and continues on physical 10. No inherited byte was accepted.

Restrained page-local apparatus

page-local note; exact one-page correspondence to authority physical 10

Authority physical 10 / printed 89. Folio 89 is recorded here and omitted from reader prose. The fixed-point proof of Proposition 1.2.2 retains the source equation $N = M + \varphi(N) + \varphi(N)M$ and the inclusion $\varphi((t)) \subset (t)^P$. Section 1.2.3 and Proposition 1.2.4 open; only condition (i) appears here, with condition (ii) continuing on physical 11. No inherited byte was accepted.

Restrained page-local apparatus

page-local note; exact one-page correspondence to authority physical 11

Authority physical 11 / printed 90. Folio 90 is recorded here and omitted from reader prose. Condition (ii) of Proposition 1.2.4, the stable-rank decomposition, the block matrices, and the footnote were retained. In particular, the authority change-of-basis matrix has x in the lower-left block, and the fixed-point expression is $f(x) = b\,a^{-1} - xcx^{(p)}a^{-1} + d\,x^{(p)}a^{-1}$. The page ends mid-sentence after the basis of H_0 . The inherited modern reflow transposes the block matrix and changes the fixed-point formula; those readings were rejected. Accepted inherited bytes: 0.

Restrained page-local apparatus

page-local note; exact one-page correspondence to authority physical 12

Authority physical 12 / printed 91. Folio 91 is recorded here and omitted from reader prose. The lower-left block change-of-basis matrix, the unit sub-F-crystal $U = \oplus Ae_i$, the intersection of images of F^n , Remark 1.2.5, and the opening of 1.3 were checked directly. Formulas (1.3.1.1)–(1.3.1.3) and the adjunction statement are present. The inherited reflow’s transposed block convention was not accepted. Accepted inherited bytes: 0.

Restrained page-local apparatus

page-local note; exact one-page correspondence to authority physical 13

Authority physical 13 / printed 92. Folio 92 is recorded here and omitted from reader prose. The page continues 1.3.1 with the Hodge and conjugate filtrations, formula (1.3.1.4), the Mazur-Ogus spectral-sequence comparison, the level condition, and the graded objects in (1.3.1.5). Super-script/subscript placement in Fil and gr, the E_1/E_2 indices, and the terminal Hodge-number citation were checked directly against the authority image. No inherited byte was accepted.

Restrained page-local apparatus

page-local note; exact one-page correspondence to authority physical 14

Authority physical 14 / printed 93. Folio 93 is omitted from reader prose. Formula (1.3.1.6), Proposition 1.3.2 with its four source-labelled conditions (*i*), (*i'*), (ii), and (iii), filtration (1.3.2.1), and Definition 1.3.3 are retained. The authority states that the Tate twist ($-i$) replaces F by $p^i F$. The proof begins and stops mid-sentence after “induit.” No candidate auto-numbering or theorem restyling was accepted.

Restrained page-local apparatus

page-local note; exact one-page correspondence to authority physical 15

Authority physical 15 / printed 94. Folio 94 is omitted from reader prose. The page continues the proof of Proposition 1.3.2, preserving formulae (1.3.3.1)–(1.3.3.3), the stable-rank inequalities and equality, the semisimple/nilpotent decomposition, and the exact sequence with the unit sub-F-crystal U_0 . The page ends grammatically open on “on,” and no closure was manufactured.

Restrained page-local apparatus

page-local note; exact one-page correspondence to authority physical 16

Authority physical 16 / printed 95. Folio 95 is omitted from reader prose. The page closes the proof of Proposition 1.3.2 and opens Remark 1.3.4. The two quotient/isomorphism statements preserve subscript Fil_i in the first pair and superscript Fil^{i+1} in the second. The inductive W_i and U_i filtrations and formula (1.3.4.1) were checked at source scale. The sentence defining V_j continues on physical 17.

Restrained page-local apparatus

page-local note; exact one-page correspondence to authority physical 17

Authority physical 17 / printed 96. Folio 96 is omitted from reader prose. The page completes Remark 1.3.4, defines a Hodge filtration and Hodge F-crystal in 1.3.5, records Griffiths transversality (1.3.5.1), and states Proposition 1.3.6. Formula (1.3.6.1) is cut after the definition of H^i and continues on physical 18. Source result classes and page boundary were preserved.

Restrained page-local apparatus

page-local note; exact one-page correspondence to authority physical 18

Authority physical 18 / printed 97. Folio 97 is omitted from reader prose. Proposition 1.3.6 is completed, the conjugate spectral sequence $(*)_r$ is retained, and §1.4 opens with formulas (1.4.1.1)–(1.4.1.3). Théorème 1.4.2 begins but its conditions are deferred to physical 19; no theorem closure was inferred.

Restrained page-local apparatus

page-local note; exact one-page correspondence to authority physical 19

Authority physical 19 / printed 98. Folio 98 is omitted from reader prose. The page contains the nine-item formula sequence (1.4.2.1)–(1.4.2.9), including closed η_{ij} , normalized τ_{ij} , and $q_{ij} = \exp(\tau_{ij})$, followed by the proof of part (i). Indices, bounds, Frobenius pullbacks, and the p-multipliers were checked directly. The page closes part (i) without entering the proof of part (ii).

Restrained page-local apparatus

page-local note; exact one-page correspondence to authority physical 20

Authority physical 20 / printed 99. Folio 99 is omitted from reader prose. The proof of Théorème 1.4.2(ii) retains formulas (1.4.2.10)-(1.4.2.13), the Frobenius automorphism $x \mapsto x^\sigma$, the normalization for $\varphi(t_i) = t_i^p$, and the operator $Vx = pF^{-1}x = px^{\sigma^{-1}}$. The source then changes from φ to an arbitrary ψ . The page ends before the displayed comparison obtained after applying e^* .

Restrained page-local apparatus

page-local note; exact one-page correspondence to authority physical 21

Authority physical 21 / printed 100. Folio 100 is omitted from reader prose. The page compares the φ and ψ specializations, then writes an arbitrary ψ as $\psi_o + pw$. The starred and double-starred displays retain the divided-power factor $p^{[n]}$, the componentwise Frobenius on w , and the source order $F(\psi_o)\psi_o^*(\nabla(D)^nb_i)$. The page ends with the constant-term identity used on physical 22.

Restrained page-local apparatus

page-local note; exact one-page correspondence to authority physical 22

Authority physical 22 / printed 101. Folio 101 is omitted from reader prose. The Taylor-expansion close of (1.4.2.13), Dwork’s Lemme 1.4.3, its two conditions, the induction on total degree, and the displayed truncated-series computations are retained. The footnote to Hazewinkel remains on this page. The authority uses LEMME and the English source word “Lemma 1” inside the citation.

Restrained page-local apparatus

page-local note; exact one-page correspondence to authority physical 23

Authority physical 23 / printed 102. Folio 102 is omitted from reader prose. The page completes Lemme 1.4.3, gives Corollaires 1.4.4 and 1.4.5, closes the proof of Théorème 1.4.2, and opens 1.4.6. The exponential integrality conditions, $p \neq 2$ hypothesis, and constant-term congruences were retained. The final A-linear homomorphism is displayed only on physical 24.

Restrained page-local apparatus

page-local note; exact one-page correspondence to authority physical 24

Authority physical 24 / printed 103. Folio 103 is omitted from reader prose. Formulas (1.4.6.1)–(1.4.6.3) and Corollaire 1.4.7 are retained. The literal session prompt calls the result “Proposition 1.4.7,” but the controlling authority clearly prints COROLLAIRE 1.4.7; the authority class controls. The regular-parameter statement, Frobenius rule (1.4.7.1), matrix-unit definition of D_{ij} , and terminal gr ∇ display continue to physical 25. No source closure was inferred.

Restrained page-local apparatus

page-local note; exact one-page correspondence to authority physical 25

Authority physical 25 / printed 104. Folio 104 is omitted from reader prose. The page completes the proof of authority Corollaire 1.4.7, gives 1.4.8 and Corollaire 1.4.9 with formula (1.4.9.1), and ends in the definition of the modules L_i . The authority prints an uppercase N for the level but a lowercase n in the condition $\text{Fil}^i H_0 = 0$ for $i \geq n+1$; it also prints $0 \leq i \leq h_m$ in the final bound of (1.4.9.1). Both are retained without emendation. The result class remains COROLLAIRE, notwithstanding the P05 prompt’s inherited wording “Proposition 1.4.7.”

Restrained page-local apparatus

page-local note; exact one-page correspondence to authority physical 26

Authority physical 26 / printed 105. Folio 105 is omitted from reader prose. The recursive definition of L_i and the limitation in level ≥ 2 close Section 1. The page then opens 2. APPLICATIONS GÉOMÉTRIQUES, 2.1 Coordonnées canoniques, A) Variétés abéliennes, and 2.1.1. The Hodge and conjugate spectral-sequence assertions continue to physical 27. No separate figure occurs.

Restrained page-local apparatus

page-local note; exact one-page correspondence to authority physical 27

Authority physical 27 / printed 106. Folio 106 is omitted from reader prose. Sections 2.1.2 and 2.1.3 are retained through the unfinished phrase “Il n’est pas difficile de montrer - nous”. In the bounds following $\tau_{ij} \in K[[t]]$, the authority literally repeats j : ($1 \leq j \leq g, 1 \leq j \leq g$). The repetition is reproduced in both source and English layers rather than silently corrected to an i, j pair. Comparator and candidate reflows were not used as source text.

Restrained page-local apparatus

page-local note; exact one-page correspondence to authority physical 28

Authority physical 28 / printed 107. Folio 107 is omitted from reader prose. The page closes 2.1.3, gives 2.1.4 Variante, opens *B*) Surfaces K3 and 2.1.5, and ends after “le”. Direct pixel inspection confirms the source inequality $g \geq 1$. The authority also prints the cross-reference (1.2.4 (i)); it is retained even though one candidate reflow silently redirects it to 1.4.2. No editorial correction is introduced.

Restrained page-local apparatus

page-local note; exact one-page correspondence to authority physical 29

Authority physical 29 / printed 108. Folio 108 is omitted from reader prose. The page gives the Hodge F -crystal, cup-product duality, Frobenius compatibility, orthogonality relation, and the opening of 2.1.6 for ordinary $K3$ surfaces. It ends after “une filtration par des sous- F -cristaux”, with the filtration itself deferred to physical 30. All displayed mathematics is preserved; there is no independent diagram.

Restrained page-local apparatus

page-local note; exact one-page correspondence to authority physical 30

Authority physical 30 / printed 109. Folio 109 is omitted from reader prose. The page completes the ordinary-filtration setup of 2.1.6 and opens Théorème 2.1.7, including parts (i), (ii), and the opening clause of (iii). It ends exactly after “alors”; the Frobenius formulas occur on physical 31 and no closure has been manufactured. The decomposition, dual basis, cup-product notation, and page boundary were retained.

Restrained page-local apparatus

page-local note; exact one-page correspondence to authority physical 31

Authority physical 31 / printed 110. Folio 110 is excluded from reader prose. The page begins with the three Frobenius formulas completing Théorème 2.1.7, carries its proof through the cup-product calculation, and opens 2.1.8 with the definitions of P_0 , P_1 , and P_2 . The page-local boundary is retained: the explanation of $M_{F=a}$ begins on physical 32. The dual basis notation $b_i \vee$ and the p^2 factor are preserved.

Restrained page-local apparatus

page-local note; exact one-page correspondence to authority physical 32

Authority physical 32 / printed 111. Folio 111 is excluded from reader prose. The page gives (2.1.8.2)–(2.1.8.5), Proposition 2.1.9, and the opening of 2.1.10. In the final paragraph the authority typesets a definition sign with “dfn” over an equals sign before “facteur direct de rang 1”; the checked transcription represents this as $\stackrel{\text{dfn}}{=}$, and the English layer renders its function as “by definition.” No silent normalization of the source notation or page boundary was made.

Restrained page-local apparatus

page-local note; exact one-page correspondence to authority physical 33

Authority physical 33 / printed 112. Folio 112 is excluded from reader prose. Théorème 2.1.11, its torsor proof, isotropic-line calculation, and the opening of Remarque 2.1.12 are complete on this page. The formulas $\text{Hom}(L_0, L_0 \perp / L_0)$, $H \otimes W_{n+2} = L_{n+1} \oplus M_{n+1}$, $x + p^{n+1}y$, and $2\langle x, y \rangle = 0 \bmod p$ were checked against the authority. The remark ends inside the phrase “cristal extension”; “universelle) [14]” remains on physical 34.

Restrained page-local apparatus

page-local note; exact one-page correspondence to authority physical 34

Authority physical 34 / printed 113. Folio 113 is excluded from reader prose. The page closes Remarque 2.1.12, proves Proposition 2.1.9, and opens Lemme 2.1.13 through the differential identity for b'_i . Direct pixel inspection confirms that the displayed Frobenius chain literally prints the unprimed a in its second term: $a' = F(\varphi')\varphi' * a' = F(\varphi)\varphi^* a = \dots$. That authority reading is retained without emendation in both source and English layers; candidate reflows regularize it and therefore do not control.

Restrained page-local apparatus

page-local note; exact one-page correspondence to authority physical 35

Authority physical 35 / printed 114. Folio 114 is excluded from reader prose. The page completes Lemme 2.1.13 and opens Théorème 2.1.14. The index orientation β_{ji}/α in $q'_i = C \prod q_j^{(\beta_{ji}/\alpha)}$, the σ – invariance argument, the formal-torus character group $\text{Hom}_{\mathbb{Z}_p}(P_0, P_1)$, and the map f are retained. The proof continues to physical 36; no commutative square has been imported onto this page.

Restrained page-local apparatus

page-local note; exact one-page correspondence to authority physical 36

Authority physical 36 / printed 115. Folio 115 is excluded from reader prose. The page contains the two commutative squares completing the proof of Théorème 2.1.14, then Définition 2.1.15 and Problèmes 2.1.16. The first square has identity on S and vertical f_W ; the second has the β/α arrow directed upward and identity on $\text{Hom}(P_0, P_1)$. The page ends exactly at the source word break “quadra-”; no completion from physical 37 is imported. The separate footnote “(ajoutée en janvier 1981) ...” crediting Nygaard is retained on this page and translated separately.

Restrained page-local apparatus

page-local note; exact one-page correspondence to authority physical 37

Authority physical 37 / printed 116. Folio 116 is excluded from reader prose. The page begins with the literal continuation “tique” of physical 36’s terminal “quadra-”, thereby completing the discussion in Problèmes 2.1.16 without silently repairing the page break. It then opens §2.2 and §2.2.1, preserves $T = \Sigma(L_0)$, formulas (2.2.1.1)-(2.2.1.3), the $\mathbb{Z}_p$ coefficients x_i , and the separate base-change footnote. The page ends at the colon introducing Théorème 2.2.2; no theorem text from physical 38 is imported.

Restrained page-local apparatus

page-local note; exact one-page correspondence to authority physical 38

Authority physical 38 / printed 117. Folio 117 is excluded from reader prose. Théorème 2.2.2 is retained with the product exponent x_i and the character-kernel formulation $T = \text{Ker}(x)$. Section 2.2.3 preserves the underlined horizontal section $\underline{x}$, both sums in (2.2.3.1), and the source’s quoted “i.e.”. Section 2.2.4 opens with the closed immersion $i : Y_0 \hookrightarrow Y$ and the Picard-class notation $c\ell(E_0)$; its exact sequence belongs to physical 39 and is not pulled backward.

Restrained page-local apparatus

page-local note; exact one-page correspondence to authority physical 39

Authority physical 39 / printed 118. Folio 118 is excluded from reader prose. The four numbered formulas (2.2.4.1)–(2.2.4.4), the multiplicative de Rham arrows $d\log$ and d , the literal terminal $\tilde{\mathcal{O}}_{Y_0}^*$ in (2.2.4.3), and the two obstruction notations ω^* , c^* are retained. In §2.2.5 the square-zero identification, the displayed definition of $I\Omega_{Y/Z}^*$, and both classes in (2.2.5.1) are kept distinct; no crystalline specialization from physical 40 is imported.

Restrained page-local apparatus

page-local note; exact one-page correspondence to authority physical 40

Authority physical 40 / printed 119. Folio 119 is excluded from reader prose. The PD-scheme/P-adic alternatives, the trivial divided powers γ , the crystalline ideal $J_{Y_0/Z}$, and formula (2.2.5.2) are retained. Authority prose immediately before (2.2.5.2) literally says “la classe de Chern cristalline de L_0 ,” while the displayed formula uses $c_1(E_0)$; the source-language and English layers preserve that L_0/E_0 discrepancy without emendation. The Cartesian diagram (2.2.5.3) has the horizontal chains $Y_0 \rightarrow Y \rightarrow X$ and $Z_0 \rightarrow Z \rightarrow S$ with vertical arrows f_0, f , and the unlabeled right-hand arrow. The page ends mid-sentence at “D’après ce”; the English layer correspondingly ends at “From what has”.

Restrained page-local apparatus

page-local note; exact one-page correspondence to authority physical 41

Authority physical 41 / printed 120. Folio 120 is excluded from reader prose. The opening completes physical 40's sentence. Lemme 2.2.6 is retained with both alternatives $p \neq 2$ and $p\mathcal{O}_Z = 0$, the canonical map to $H^2(Y, \mathcal{O}_Y)$, and the Fil^1 criterion. Section 2.2.7 distinguishes the absolute class (2.2.7.1), its crystalline direct-image section (2.2.7.2), and the value on a PD-thickening (2.2.7.3). The final displayed section continues syntactically onto physical 42.

Restrained page-local apparatus

page-local note; exact one-page correspondence to authority physical 42

Authority physical 42 / printed 121. Folio 121 is excluded from reader prose. The page completes the section-valued Chern-class sentence from physical 41, preserves the factorization (2.2.7.4), the notation $\widehat{Z}' = D_{Z_0}(Z')$, and the Cartesian diagram (2.2.7.5) with all three vertical arrows. Section 2.2.8 opens with the double-angle notation $W\langle\langle t \rangle\rangle$ and the divided-factorial series $\sum a_i t^i / i!$. The page ends at “Considérons maintenant”; the ideal considered is confined to physical 43.

Restrained page-local apparatus

page-local note; exact one-page correspondence to authority physical 43

Authority physical 43 / printed 122. Folio 122 is excluded from reader prose. The page begins by completing physical 42's "Considérons maintenant". Formulas (2.2.8.2)–(2.2.8.3) retain the multi-index decomposition $m = rq+s$, the divided-power notation $(t^r)^{[q]}$, and the factor $(m!/q!)t^{[m]}$. The proof of 2.2.2 opens as §2.2.9 with p^m the exact p-divisibility of $c_1(L_0)$, $p \nmid y$, and T' defined by $q_1^{p^m} - 1$. Step a ends after the conjunction "et"; its consequence begins on physical 44.

Restrained page-local apparatus

page-local note; exact one-page correspondence to authority physical 44

Authority physical 44 / printed 123. Folio 123 is excluded from reader prose. Step a is continued with the bound of order $\leq p^m - 1$, the schemes $Y_n = \operatorname{Spec}(k[t_1]/t_1^n)$, and the PD-envelope D_n . The authority diagram includes the upper and lower horizontal arrows, two vertical arrows, and a diagonal arrow from $\operatorname{Spec}(k)$ toward $\operatorname{Spf}(W[[t_1]])$; that topology is retained. The two horizontal Chern-class displays preserve $p^m \log q_1$ versus $\log q_1^{p^m}$. The page ends at the second display; the following parenthetical begins on physical 45.

Restrained page-local apparatus

page-local note; exact one-page correspondence to authority physical 45

Authority physical 45 / printed 124. Folio 124 is excluded from reader prose. The opening parenthetical belongs to the formula frozen on physical 44. The universal PD map retains the divided-power element $x^{[i]}$, the distinction $i = 1$ versus $i \geq 2$, and the modulus (p, t_1^{n+1}) . Step b opens with Z and Z_n , then displays a four-line equality of PD-envelopes; the successive ideals $(p^n, q_1^{p^m} - 1)$, $(p, q_1^{p^m} - 1)$, $(p, (q_1 - 1)^{p^m})$, and $(t_1^{p^m})$ are not collapsed. The page ends at “que”.

Restrained page-local apparatus

page-local note; exact one-page correspondence to authority physical 46

Authority physical 46 / printed 125. Folio 125 is excluded from reader prose. The page completes step *b* with the two-line Chern-class display and the specialization $q_1^{p^m} - 1 = 0$ on Z_{n+1} . Step *c* retains the full multi-index $n = (n_2, \dots, n_{20})$, T'_n , and the increment notation $n + 1_i$. Step *d* opens the maximality argument for T' . The final words are exactly “Or, le même”; the calculation and bibliography remain on physical 47.

Restrained page-local apparatus

page-local note; exact one-page correspondence to authority physical 47

Authority physical 47 / printed 126. Folio 126 is excluded from reader prose. The opening display completes the maximality argument and the proof of Théorème 2.2.2. Bibliographie begins on the same page and contains items [1]–[13]. Bibliographic titles, author punctuation, journal abbreviations, page ranges, and source-language forms are retained as citation matter in both readers rather than silently translated or normalized. Items [14]–[17] are confined to physical 48.

Restrained page-local apparatus

page-local note; exact one-page correspondence to authority physical 48

Authority physical 48 / printed 127. Folio 127 is excluded from reader prose. The page contains Bibliographie [14]–[17], followed by two spatially separate contributor address blocks for *P. Deligne* and *L. Illusie*. The institutional wording, accents, building number, postal codes, and BURES/YVETTE slash are retained. The substantial lower-page blank remainder is preserved by the authority facsimile and recorded here rather than filled. This page closes the French main exposition; Nicholas *M. Katz*’s separately authored English appendix begins only on physical 49.

Restrained page-local apparatus

page-local note; exact one-page correspondence to authority physical 49

Authority physical 49 / printed 128. No printed folio is visible. This page is the explicit source-language and authorship boundary: the French Deligne exposition has ended, and APPENDIX TO EXPOSE V begins in English under the separate byline Nicholas M. Katz. The standalone English reader therefore performs SOURCE_ALREADY_ENGLISH_REPLAY, not translation. A1 and Uniqueness Lemma A1.1 are preserved with the commutative square and all Φ , σ , and Frobenius labels.

Restrained page-local apparatus

page-local note; exact one-page correspondence to authority physical 50

Authority physical 50 / printed 129. Folio 129 is excluded from reader prose. The proof of A1.1 preserves the finite-flat $\text{Ker}(\Phi)$ argument, Fontaine’s Dieudonné-module conditions a) and b), and the p -labelled commutative square. The final line literally uses the exponent h_i in $'(\widehat{G}_m)^{\text{hi}}'$ after the preceding diagram uses n_1 and n_2 ; this authority reading is retained without normalization. The page ends after the word 'our'.

Restrained page-local apparatus

page-local note; exact one-page correspondence to authority physical 51

Authority physical 51 / printed 130. Folio 130 is excluded from reader prose. The p-power iteration, logarithmic linearization through $\widehat{G}_a$, Q_p/Z_p coefficient argument, and Q.E.D. closing A1.1 are retained. Corollary A1.2 begins and its character-group isomorphism is complete on this page. The English reader replays the already-English source.

Restrained page-local apparatus

page-local note; exact one-page correspondence to authority physical 52

Authority physical 52 / printed 131. Folio 131 is excluded from reader prose. Corollary A1.2(2)-(5) is retained through condition c), including Φ – pullback formulas for q , ω , and τ ; the path-integral exponential; the completed tensor product; and the formal spectrum $W[[q_1 - 1, \dots, q_n - 1]]$. The page-local ending after the continuous-differentials condition is preserved.

Restrained page-local apparatus

page-local note; exact one-page correspondence to authority physical 53

Authority physical 53 / printed 132. Folio 132 is excluded from reader prose. Conditions d) and e) and the proof of Corollary A1.2 are retained, including the toroidal identification, coefficient functions, product character, and Dieudonné-Dwork integrality display. The authority's literal grammatical sequence 'is integral (i.e. lies in A) and lies $q(O) = 1'$ is preserved rather than silently repaired. The page ends at $\Phi^*(\tau^{(\sigma)}) = p\tau$.

Restrained page-local apparatus

page-local note; exact one-page correspondence to authority physical 54

Authority physical 54 / printed 133. Folio 133 is excluded from reader prose. The proof of Corollary A1.2 closes with Q.E.D.; its literal ' $(\widehat{G}_m)^r$ ' is retained. A2, SHARPENINGS OF 1.4 and Corollary A2.1 are then reproduced in full, including the three brace-formula groups, the source word 'formulaire', and the literal comma in $\tau_{i,j}$ within the pullback formula. Physical 55 begins Further Corollary A2.2; none of that material is imported here.

Restrained page-local apparatus

page-local note; exact one-page correspondence to authority physical 55

Authority physical 55 / printed 134. Folio 134 is excluded from reader prose. Further Corollary A2.2 and its proof are reproduced in full. The heading's opening parenthesis before 'Analogue' has no corresponding closing parenthesis before 'Given' in the authority and is not supplied. Source forms 'formulaire', 'tel quel', 'resulte', and 'definible' are retained. In the terminal formula the authority prints $\Phi_{\text{can}}(q_{ij}^{(\sigma)}) = (q_{ij})^P$ without a pullback star; this is preserved and not harmonized with physical 58.

Restrained page-local apparatus

page-local note; exact one-page correspondence to authority physical 56

Authority physical 56 / printed 135. Folio 135 is excluded from reader prose. A3 opens under Nicholas *M.* Katz's appendix authorship. The canonical-subgroup paragraph, the map $F_{\text{can}} : X \rightarrow X/H_{\text{can}}$, the classifying map $\Phi_{\text{can}} : M \rightarrow M^{(\sigma)}$, the isomorphism $\Phi_{\text{can}}^*(X^{(\sigma)}) \cong X/H_{\text{can}}$, the two-arrow diagram to M , and the induced H_{DR}^1 map are retained. The page ends exactly after 'Because F_{can} is' and is continued on physical 57.

Restrained page-local apparatus

page-local note; exact one-page correspondence to authority physical 57

Authority physical 57 / printed 136. Folio 136 is excluded from reader prose. The page begins with the literal continuation 'a physical morphism'. Theorem A3.1, its Ext description, product-torus isomorphism, proof, and Q.E.D. are complete. A4 and Corollary A4.1 then open. In the two-line Hodge-filtration brace, the authority prints $(\text{Fil}^2)^{(\sigma)}$ on the left in both lines; the second line is not normalized to Fil^1 . Physical 58 contains the proof.

Restrained page-local apparatus

page-local note; exact one-page correspondence to authority physical 58

Authority physical 58 / printed 137. Folio 137 is excluded from reader prose. Corollary A4.1 is completed with the authoritative identity $\Phi_{\text{can}}^*(q_i^{(\sigma)}) = (q_i)^P$ for $i = 1, \dots, 20$, including the pullback star that is absent from the terminal formula on physical 55. Q.E.D., the substantial blank remainder, and true end of file are preserved; no material follows.